AGENTIC AI KNOWLEDGE BASE • WEATHER & PLAY KNOWLEDGE GRAPH

Weather Parameters & Activity Decision Matrix

1. Knowledge Base Overview & Intent

This document serves as an explicit Retrieval-Augmented Generation (RAG) context source for the Weather & Play Agentic AI system. The application evaluates eight core meteorological parameters—**temperature, is_day, wind_speed, rain, snowfall, relative_humidity, and precipitation**—to recommend optimal indoor and outdoor activities for users based on query parameters.

Key Meteorological Evaluation Parameters

- **temperature (°C):** Thermal comfort threshold classification (<0°C Freezing, 0-10°C Cold, 11-20°C Mild, 21-30°C Warm, >30°C Hot).
- **is_day (1 / 0):** Solar illumination flag (1 = Daylight, 0 = Night). Restricts outdoor visibility-dependent activities.
- **wind_speed (km/h):** Aerial stability (<15 Calm, 15-30 Moderate, >30 High/Dangerous for projectile sports).
- **rain / snowfall / precipitation (mm/h):** Active precipitation indicators dictating indoor transition rules.
- **relative_humidity (%):** Evaporative cooling/heat index metric (>80% increases thermal stress).

2. Rule-Based Classification Matrix

Condition Profile	Temp (°C)	is_day	Wind (km/h)	Precip / Rain / Snow	Humidity	Activity Decision Focus
P1: Clear Warm Day	22°C - 28°C	1	< 15 km/h	0.0 mm (None)	40% - 60%	Optimal Outdoor Aerobic & Field Games
P2: Hot Humid Day	> 32°C	1	< 20 km/h	0.0 mm (None)	> 75%	Shaded/Water Sports or Air-Conditioned Indoor
P3: Heavy Rain / Storm	15°C - 22°C	1 / 0	> 25 km/h	> 5.0 mm/h (Rain)	> 85%	Strict Indoor Transition (Tabletop, Gym)
P4: Freezing Snowfall	< 0°C	1	< 20 km/h	> 2.0 mm/h (Snow)	60% - 80%	Winter Sports (Outdoor) / Hearth Games (Indoor)
P5: Clear Cool Night	10°C - 16°C	0	< 10 km/h	0.0 mm (None)	50% - 70%	Low-light Outdoor Walks or Indoor Social Games
P6: High Wind Day	18°C - 24°C	1	> 35 km/h	0.0 mm (None)	45% - 65%	Wind-assisted Outdoor (Kites) / Indoor Field

3. Detailed Weather Profiles & Activity Mappings

Profile A: Ideal Mild Day (Clear Skies & Low Wind)

TEMPERATURE	DAYLIGHT (IS_DAY)	WIND SPEED	PRECIPITATION
18°C - 25°C	**1 (Day)**	**5 - 15 km/h**	**0.0 mm**

Recommended Outdoor Games

- **Football / Soccer:** Optimal temperature prevents overheating during high cardio.
- **Tennis & Badminton:** Low wind speed allows accurate shuttlecock/ball trajectory.
- **Cycling & Running:** High endurance outdoor workouts.
- **Basketball:** Dry outdoor courts offer maximum grip.

Recommended Indoor Games

- **Squash / Badminton:** Indoor court games.
- **Table Tennis:** Quick-reaction indoor table game.
- **Board Games & Chess:** Casual social games if preference leans indoors.

Profile B: Active Rainy Condition (High Precipitation)

TEMPERATURE	DAYLIGHT (IS_DAY)	WIND SPEED	RAIN / PRECIP
12°C - 22°C	**1 or 0**	**15 - 30 km/h**	**> 4.0 mm/h**

Recommended Outdoor Games

- **Mud Rugby / Rain Football:** Only for casual play with water-resistant gear.
- *Note:* Most outdoor activities are suspended due to slipping hazards.

Recommended Indoor Games

- **Bowling:** Complete shelter from precipitation and wind.
- **Indoor Swimming:** Heated pool activity regardless of rain.
- **Billiards / Pool / Snooker:** Precision indoor tabletop games.
- **Escape Rooms / Arcade Games:** Group cognitive activities.

Profile C: Snowfall & Sub-Zero Weather

TEMPERATURE	DAYLIGHT (IS_DAY)	WIND SPEED	SNOWFALL
-5°C - 1°C	**1 (Day)**	**10 - 20 km/h**	**> 1.5 mm/h**

Recommended Outdoor Games

- **Ice Hockey & Skating:** Natural ice formation on prepped rinks.
- **Snowball Fights & Sledding:** Fresh snow cover activities.
- **Skiing / Snowboarding:** Active mountain/slope sports.

Recommended Indoor Games

- **Indoor Ice Skating Rink:** Climate-controlled winter sports.
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